I Trust It, but I Don't Know Why: Effects of Implicit Attitudes Toward Automation on Trust in an Automated System

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Objective: This study is the first to examine the influence of implicit attitudes toward automation on users' trust in automation.

Background: Past empirical work has examined explicit (conscious) influences on user level of trust in automation but has not yet measured implicit influences. We examine concurrent effects of explicit propensity to trust machines and implicit attitudes toward automation on trust in an automated system. We examine differential impacts of each under varying automation performance conditions (clearly good, ambiguous, clearly poor).

Method: Participants completed both a self-report measure of propensity to trust and an Implicit Association Test measuring implicit attitude toward automation, then performed an X-ray screening task. Automation performance was manipulated within-subjects by varying the number and obviousness of errors.

Results: Explicit propensity to trust and implicit attitude toward automation did not significantly correlate. When the automation's performance was ambiguous, implicit attitude significantly affected automation trust, and its relationship with propensity to trust was additive: Increments in either were related to increases in trust. When errors were obvious, a significant interaction between the implicit and explicit measures was found, with those high in both having higher trust.

Conclusion: Implicit attitudes have important implications for automation trust.

Application: Users may not be able to accurately report why they experience a given level of trust. To understand why users trust or fail to trust automation, measurements of implicit and explicit predictors may be necessary. Furthermore, implicit attitude toward automation might be used as a lever to effectively calibrate trust.

Keywords: automation, trust, reliance, propensity to trust, implicit, unconscious, ambiguous

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Over- and/or underreliance on automation can have life-and-death implications, creating a pressing need to understand when and why users rely on automation. Failure to use automation appropriately has contributed to tragedies such as the fatal crash of an Airbus A320 when the pilots failed to correct an autopilot navigation error (Sparaco, 1995). Research has established that user trust is associated with automation reliance, with users relying more heavily on automation that is more trusted (e.g., de Vries, Midden, & Bowhuis, 2003; Merritt, 2011; Merritt & Ilgen, 2008). Because different users may have varied levels of trust in the same automation (Merritt & Ilgen, 2008), better understanding the user individual differences that influence trust may assist researchers and practitioners in predicting or controlling user reliance behaviors.

Trust has been defined as "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability" (Lee & See, 2004, p. 51). Behavioral reliance on automation is empirically predicted by user trust in the automation (de Vries et al., 2003; Lee & Moray, 1992; Merritt, 2011; Wang, Jamieson, & Hollands, 2009); thus, either overtrust or undertrust may be problematic. Overtrust may produce automation misuse, in which users rely too heavily on automation, failing to override the system when necessary. In contrast, undertrust may produce automation disuse, in which users fail to rely on automation when doing so would improve performance (Parasuraman & Riley, 1997).

To prevent misuse and disuse, it may be necessary to calibrate user trust in the automated system so that appropriate levels of trust are experienced given the automation's reliability (Lee & See, 2004; Parasuraman & Miller, 2004). Because various users likely have different perceptions of the same automated system (Merritt & Ilgen, 2008), researchers have examined individual differences affecting trust. Several individual differences that influence trust have been empirically studied, including self-confidence (Lee & Moray, 1994), mood (Merritt, 2011), personality (Merritt & Ilgen, 2008; Szalma & Taylor, 2011), and propensity to trust machines (Merritt & Ilgen, 2008).

Past empirical work on attitudinal predictors of trust has focused predominantly on explicit predictors, rarely assessing implicit predictors. However, both explicit and implicit processes affect perceptions and behavior (Gawronski & Bodenhausen, 2006). Explicit processes are cognitively effortful and conscious. Because we are aware of them, they can be measured via self-report. In contrast, implicit processes affect behavior automatically and predominantly unconsciously, making them largely inaccessible to standard self-report measures (cf. Gawronski, Hofmann, & Wilbur, 2006). Although we lack awareness of implicit processes, they may drive a large proportion of our daily activities (Bargh & Chartrand, 1999). Individuals have been demonstrated to possess implicit attitudes toward a variety of objects, including demographic groups, product brands, and alcohol consumption (Greenwald, Poehlman, Uhlmann, & Banaji, 2009; Maison, Greenwald, & Bruin, 2004; Payne, Govorun, & Arbuckle, 2008).

The present study's focus is on implicit attitude toward automated systems, which is defined as the positivity of the individual's mental associations with the concept of automation, on a continuum ranging from very negative to very positive. When the mental concept of automation is activated, the associated evaluation (i.e., attitude) becomes automatically activated as well (e.g., Gawronski & Bodenhausen, 2006). The strength of the automatic link between "automation" and either "good" or "bad" can be described as a person's implicit attitude toward automation. In certain cases where the implicit attitude activation is consciously perceived, it can be described as a "gut reaction" (Ranganath, Smith, & Nosek, 2008). Theoretically, these gut reactions serve as the basis for subsequent judgments and behaviors. Explicit evaluations, such as explicit trust, are expected to correspond with gut reactions when perceivers lack either the ability or motivation to override them (Fazio, 1990). One reason for this correspondence is that implicit attitudes can affect the perception and interpretation of subsequent information, so that new information is perceived as consistent with the implicit attitude (Devine, 1989). Through these processes, implicit attitudes toward automation may affect trust in automation.

Implicit attitudes are often assessed via measurements of the speed with which individuals can respond to the attitude object (i.e., automation) and an evaluative representation (i.e., synonyms of the word *good*). When individuals can more quickly pair "automation" and "good" than they can pair "automation" and "bad," they possess more positive implicit attitudes toward automation. Note that this attitude reflects implicit associations related to the category of automated systems in general rather than toward any particular automated system. We expect that implicit attitude toward automation in general may have significant effects on users' interactions with a particular automated system. Specifically, the goal of the present study is to assess users' implicit attitude toward automation and its utility in predicting explicit (i.e., self-reported) trust in a specific automated system.

Dual process models suggest that perceptions of others are influenced by both implicit and explicit processes. According to such models, implicit and explicit processes are thought to be related, but qualitatively separate (Wilson, Lindsey, & Schooler, 2000). Although the process behind the formation of implicit attitudes is somewhat poorly understood, the mental associations underlying implicit attitudes are thought to develop through evaluative conditioning, where the category (e.g., "automation") is repeatedly paired with positively or negatively valenced stimuli (Olson & Fazio, 2001). For example, watching news stories in which automation effectively reduces accidents is likely to produce more positive implicit attitudes toward automation, whereas hearing stories of plane crashes caused by faulty autopilot systems is likely to produce more negative implicit attitudes toward automation. Thus, implicit attitudes toward automation may reflect the total aggregate personal and cultural experiences an individual has had (or heard about) relevant to anything falling within the category of automation.

These experiences need not be consciously remembered or endorsed. According to the associative-propositional evaluation model (Gawronski & Bodenhausen, 2006), implicit attitudes operate purely via association. Thus, they reflect the degree to which two concepts have been associated via repeated pairings but are unrelated to whether the individual consciously believes the association to be "correct." A key feature of implicit attitudes is that they may affect the individual's behavior regardless of whether the individual consciously endorses them.

In contrast, explicit attitudes operate via propositional processes in which the attitude is subjected to truth values ("true" or "false") and will be endorsed only when the individual believes the attitude to be true. Therefore, implicit and explicit attitudes toward any given object may or may not converge, depending on whether the individual's automatic associations mirror what the individual consciously believes to be true (cf. Nosek, 2005). Because implicit and explicit processes are distinct, and because they often dissociate, implicit preferences may provide predictive power that cannot be obtained via traditional explicit measures.

The process by which implicit attitudes toward automation may affect user trust is similar in some ways to Lee and See's (2004) conception of analogical processes of trust. In analogical processes, trust develops via inferences made based on category membership (i.e., "This is an automated system; automated systems are trustworthy; this system is trustworthy"). Likewise, implicit attitudes are category based in that activation of the category of automation is accompanied by automatic activation of associated evaluations. These mental associations are then hypothesized to affect trust. However, although analogical processes could occur either implicitly or explicitly (i.e., individuals could be aware of their explicit assumptions about trustworthiness of automation and the effects that those beliefs have on their actual levels of trust), the effects of implicit attitudes on outcomes have been shown to occur outside of awareness (e.g., Bargh, Chen, & Burrows, 1996). Implicit attitudes could serve as a partially mediating process in analogical trust; they could serve as one of the mechanisms by which category membership affects trust.

In addition, implicit attitude toward automation can be linked to what Lee and See describe as an "affective process." They describe affective influences on trust as sometimes automatic and outside conscious awareness—consistent with the conceptualization of implicit attitudes. In contrast, however, we characterize implicit attitude toward automation as an attitude (i.e., an association with good/bad) as opposed to an emotion (i.e., affect). Specifically, an attitude is defined as an overall evaluation of an object on a latent scale ranging from *very bad* to *very good*. In contrast, emotions are specific arousal states with labels such as *anger* or *joy*. Attitudinal evaluations are influenced by both cognitions (i.e., beliefs) and emotions toward the attitude object (e.g., Olson & Zanna, 1993; Petty, Wegener, & Fabrigar, 1997). Thus, although attitudes and affect toward an object are expected to be associated, they are considered separate constructs (e.g., Weiss, 2002).

Although a large body of research has examined the role of implicit attitudes in other domains (e.g., diversity and discrimination, health and addiction), fewer studies have examined implicit attitudes in the context of trust and automation. Some work has focused on implicit trust in regard to organizational safety culture (e.g., Burns, Mearns, & McGeorge, 2003) and implicit attitudes in regard to perceived risk from nuclear power plants (e.g., Siegrist, Keller, & Cousin, 2006). However, we found no empirical studies examining how implicit attitudes toward automation might affect users' explicit trust in automated systems.

In the human factors and ergonomics literature, one prior study has empirically examined implicit attitudes as we conceptualize them. Molesworth and Chang (2009) examined the effects of implicit attitudes toward risky flight behavior on pilots' in-flight behavior. To do so, they measured implicit attitudes using an Implicit Association Test (IAT; Greenwald,

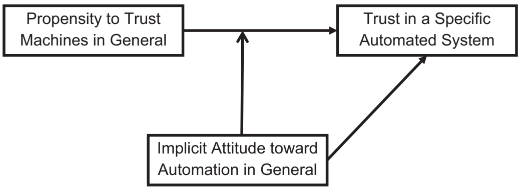

Figure 1. Hypothesized model.

McGhee, & Schwartz, 1998). They then assessed pilots' behavior in a flight simulator and found that the IAT significantly predicted risky flight behavior even when explicit measures had been accounted for.

In the present study, we examine the utility of implicit preference for automation (i.e., the individual's mental associations between the category of automation and an evaluative valence) in predicting the individual's trust in a specific automated decision aid. In doing so, we compare the effects of this implicit measure to the effects of an established explicit predictor of trust: propensity to trust machines.

Propensity to trust in general is "a stable, trait-like tendency to trust or not trust others" (Merritt & Ilgen, 2008, p. 195), and *propensity to trust machines* reflects the tendency to be trusting of machines broadly, rather than specific to any given machine. In contrast, we conceptualize trust as specific to the automated system with which participants interact. Our hypothesized model is depicted in Figure 1.

Propensity to trust machines in general and actual trust in a specific automated system have been established as separate constructs in past work (e.g., Merritt, 2011). Propensity to trust has been associated with a particular pattern of actual trust over time. Users with higher propensity to trust often show higher initial levels of trust in a specific automated system. These levels persist until automation errors are encountered, at which time individuals with higher propensity to trust often display a more marked decrease in trust than those with lower propensity to trust (Dzindolet, Pierce, Beck, & Dawe, 2002; Madhavan & Wiegmann, 2007; Merritt & Ilgen, 2008). This decrease in trust is particularly evident when automation errs on easy tasks (Madhavan, Wiegmann, & Lacson, 2006). To explain this decrease, scholars have suggested that individuals with higher propensity to trust may have a stronger perfect automation schema. This schema is a mental construct that reflects the participants' expectations for perfect automation performance (Dzindolet et al., 2002).

In most situations, schema activation facilitates the integration of expectancy-consistent information from the environment (e.g., Fiske, 1998; Fyock & Stangor, 1994; Macrae & Bodenhausen, 2000; Macrae, Bodenhausen, Milne, & Jetten, 1994). People are more likely to remember and use information that confirms their expectations, and they are also likely to interpret ambiguous information as expectancy consistent. This information processing bias tends to produce judgments in line with expectations when the target's behavior is either expectancy consistent or ambiguous. This effect may produce the higher levels of trust initially observed for individuals high in propensity to trust.

However, schema activation also increases individuals' sensitivity to unexpected information such that information blatantly opposing expectations is emphasized in forming judgments. Thus, when counterexpectant information is obvious or striking enough to overcome the processing bias discussed earlier, individuals with stronger schemas weight that information more heavily than individuals with weaker schemas (Hastie & Kumar, 1979; Srull & Wyer, 1989). The effect of this violation of expectations may explain the more marked decrease in trust after encountering automation errors for users with higher propensity to trust.

The theory discussed earlier implies that the effects of propensity to trust machines will depend on the degree to which the automation's errors are obvious to users. Expectations most often produce a tendency toward assimilation of expectancy-consistent information (Macrae & Bodenhausen, 2000), so we expect that when the automation's performance is clearly good, users with high propensity to trust will emphasize information consistent with their expectations for high performance. Thus, they will have high trust in the automated system. Similarly, when the automation's performance is ambiguous, we expect that users with higher propensity to trust may be more likely to perceive ambiguous automation performance as positive. Therefore, when performance is ambiguous, we also expect users with high propensity to trust to have higher trust in the system.

Hypothesis 1a: When automation performance is unambiguously good, propensity to trust machines will be associated with higher actual trust in the automated system.

Hypothesis 1b: When automation performance is ambiguous, propensity to trust machines will be associated with higher actual trust in the automated system.

However, we expect this trend to reverse when users encounter obvious machine errors. When schemas are violated by strongly counterexpectant information, individuals are particularly likely to notice and remember the violating information (e.g., Stangor & McMillan, 1992). Thus, when the automation makes obvious errors, we expect to see the same marked decrease in trust associated with higher propensity to trust shown in past work. We therefore hypothesize,

Hypothesis 1c: When the automation makes obvious errors, propensity to trust machines will be associated with lower actual trust in the automated system.

Consistent with dual process models, implicit preference for automation is expected to have either an additive or interacting relationship with propensity to trust in predicting actual trust. Because researchers have not yet identified means for predicting when an additive versus an interactive relationship might occur, we test both. Implicit processes may have their largest effects in ambiguous situations (Dovidio & Gaertner, 2000; Fiske, 1998), suggesting that implicit preference for automation may have its strongest main effects on trust when automation performance is ambiguous.

Research question: Does implicit preference for automation significantly predict actual trust when the machine's performance is clearly good, ambiguous, and clearly negative?

METHOD

This study was an online, three-condition, within-subjects design in which each participant encountered three levels of automation performance (clearly good, ambiguous, and clearly poor). The sample was 69 college students (age M = 25; 77% female; 58% White). All participants completed the study via the Internet. Note that because participants accessed the study from a variety of locations, specific location effects are not expected to have played a systematic role in the results found.

Participants first completed the explicit (i.e., self-report) propensity to trust measure and the implicit attitude toward automation measure described in more detail later. Next, they completed a 30-trial X-ray screening task similar to that used in past research (Merritt, 2011). In this task, participants viewed X-ray images of luggage with the goal of determining whether each image contained a gun or knife. While completing the task, they were placed under cognitive load (i.e., asked to remember an eight-digit number; Gilbert & Hixon, 1991) to more accurately simulate real-world working conditions with multiple simultaneous demands. For each image, they received advice from a (fictitious) automated system that we called the Automated Baggage Inspector (ABI). On each slide, participants first provided their initial, unaided decision, next received the ABI's advice, and then made a final decision. No performance feedback was provided. The 30 slides were divided into three blocks of 10.

Automation Performance Manipulation

Automation performance (clearly good, ambiguous, and clearly poor) was manipulated within subjects by varying both the number of errors made and the obviousness of those errors. For the clearly good condition (first 10 slides), the ABI made no errors. Furthermore, the slides selected for this block were quite easy, and when a weapon was present, it was displayed in an obvious manner (see Figure 2).

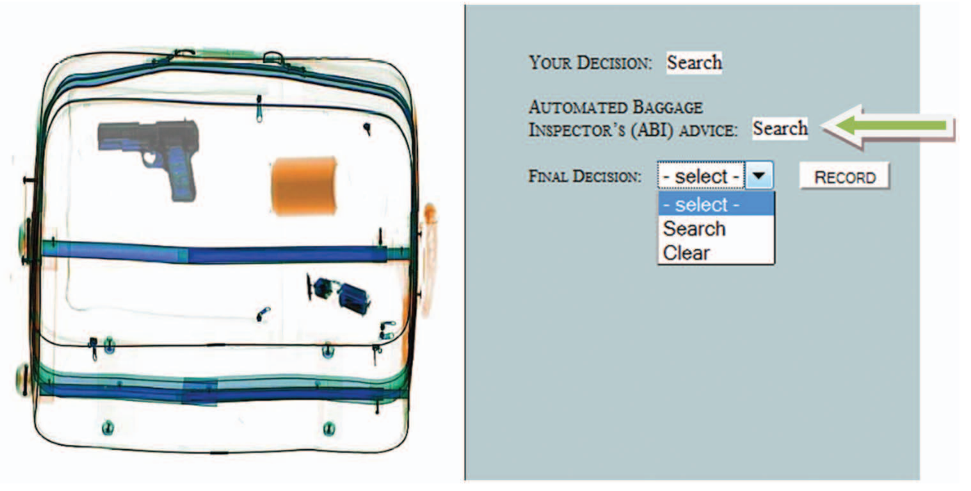

Figure 2. Example X-ray task slide in the "automation performance clearly good" condition.

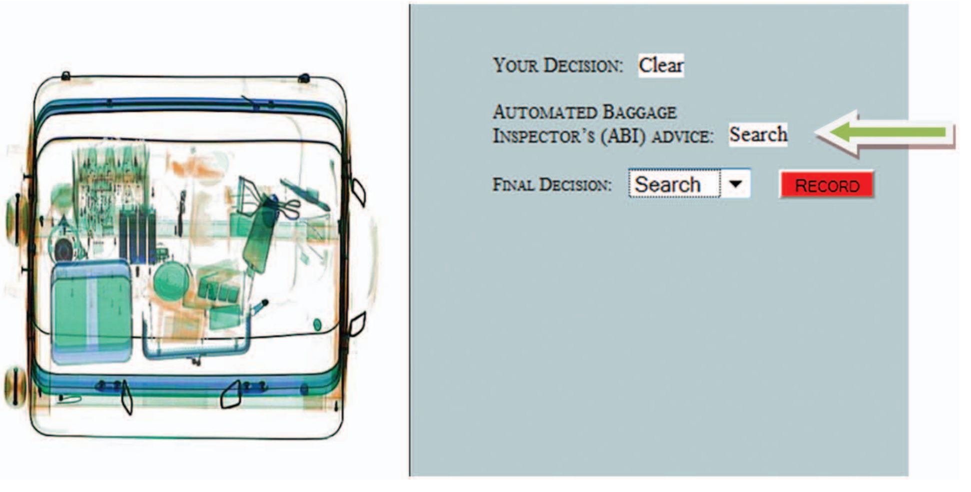

Figure 3. Example X-ray task slide in the "automation performance ambiguous" condition.

In the ambiguous condition (Slides 11–20), the difficulty of the slides increased. Each slide contained several items, and when weapons were present, they were partially obscured and/or presented at an angle that disguised their identity (see Figure 3). In this condition, the automation made two errors (one miss, one false alarm), but errors were not obvious.

Finally, in the clearly poor condition (Slides 21–30), the ABI also made two errors (one miss, one false alarm), but these were intended to be obvious to participants (see Figure 4). Each of these errors occurred on extremely easy slides, which has been found in past work to produce large decreases in trust (Madhavan et al., 2006).

As a manipulation check, we calculated participants' accuracy on their initial (i.e., before receiving automation advice) decisions for each slide. The results are reported in Table 1. As expected, participants' unaided performance was significantly higher in the clearly good and clearly poor performance conditions than in the ambiguous condition ($t = 9.48$ and 9.81, respectively). Unaided performance in the clearly good condition and that in the clearly poor

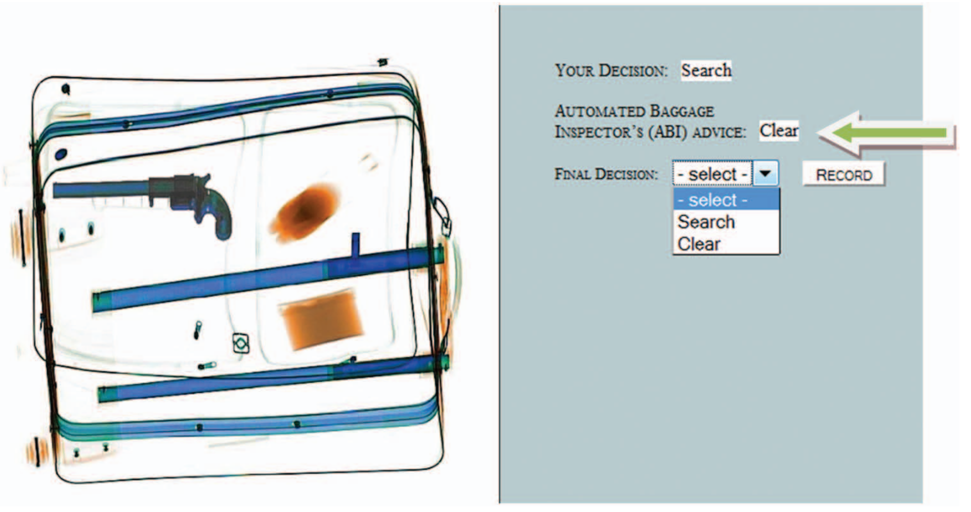

Figure 4. Example X-ray task slide in the "automation makes obvious errors" condition.

TABLE 1: Manipulation Check: Mean Unassisted Performance by Condition

Condition	*M*	*SD*
Clearly good	0.82*	0.16
Ambiguous	0.61	0.13
Clearly poor	0.86*	0.13

*Significantly different from ambiguous condition, $p < .05$.

condition did not significantly differ from each other ($t = .94$, $p = .35$). These results indicate that participants were able to perform the task significantly better without assistance in Blocks 1 and 3 than in Block 2. This supports the assertion that the participants were better able to evaluate the automation's performance in these blocks because when people can ascertain the correct response on their own, they can more accurately judge whether the automation has also reached the correct decision. When individuals do not know the correct response, they are less able to accurately judge the automation's performance.

In addition, we examined the rates of agreement with the system for trials on which the automation provided incorrect responses. Because the ABI was always correct in Block 1, we focused on the rates of agreement with the automation in Blocks 2 and 3. In Block 2, rates of agreement with incorrect advice from the automation were significantly higher than rates of agreement in Block 3 for both the false alarm ($t = 5.84$, $p < .01$) and the miss ($t = 41.13$, $p < .01$). This suggests that automation errors were significantly more apparent to participants in Block 3 than in Block 2.

Measures

The dependent variable, actual trust in the automated screening aid, was assessed using a six-item self-report scale employed by Merritt (2011). The item responses were on a 5-point Likert-type scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). The scale was administered following each of the three task blocks. Example items include "I trust the ABI" and "I have confidence in the advice given by the ABI." The alpha internal consistencies for the three task blocks were .89, .91, and .96, respectively.

Propensity to trust machines was measured with the six-item scale used by Merritt (2011). The response options were on a 5-point Likert-type scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). The items can be found in the appendix. Scale internal consistency alpha was .87.

Because implicit attitudes are theorized to often operate outside of awareness, they are typically assessed using measures based on reaction times. To measure implicit attitude toward automation, an IAT (Greenwald et al., 1998) was created. Of the implicit attitude measures, the IAT has perhaps the most validity evidence as well as the strongest internal consistency (Nosek, Greenwald, & Banaji, 2005). Example IATs are available for demonstration through Project Implicit (IAT Corp, 2011). Our IAT was created using Inquisit by Millisecond Software (Inquisit, 2011). The IAT assesses the relative strengths of evaluative associations with a focal category (in this case, automation) and a contrasting category (in this case, humans).

During the IAT, participants were asked to categorize good words (e.g., *marvelous, superb*), bad words (e.g., *tragic, horrible*), words representing humans (*human* and *person*), and those representing automation (*automation* and *machine*) into their superordinate categories (i.e., good/bad or automation/human). In creating an IAT, it is important that stimulus words can be easily distinguished and sorted into only one category. It has been suggested to avoid stimuli that are only tangentially related to the categories (Nosek et al., 2005). Focus groups identified two strongly related words for the human category (*human* and *person*) and the automation category (*automation* and *machine*) by consulting a thesaurus and generating synonyms. Options judged as not fully reflecting the desired categories were eliminated by group consensus. Two stimulus words per category were selected in light of findings suggesting lower validity for IATs that use only one stimulus per category (Nosek et al., 2005). The evaluative category (i.e., good/bad) words were adopted from Project Implicit's race IAT (words used: *joy, love, peace, wonderful, pleasure, glorious, laughter, happy; agony, terrible, horrible, nasty, evil, awful, failure, hurt*).

Following a number of practice trials during which the participants were acquainted with the sorting task, test blocks were completed in which the categories (human or automation) were paired together with the evaluations. The participant was asked to provide the same response to a combination of category and evaluation ("press 'e' if the word is associated with automation *or* good; press 'i' if the word is associated with human *or* bad"). In subsequent blocks, the pairing was reversed ("press 'e' if the word falls into the categories human *or* good; press 'i' if the word falls into the categories automation *or* bad"). See Figure 5 for a screenshot of our IAT task.

Figure 5. Screenshot from Implicit Association Test measuring implicit preference for automation.

IATs use response latencies to measure implicit associations, with shorter response latencies representing stronger associations and thus a stronger preference (Greenwald & Banaji, 1995; Greenwald et al., 1998). Therefore, the more positive an individual's implicit attitude toward automation, the more quickly he or she should be able to complete the task when "automation" and "good" are paired together, and the more difficulty he or she should have when "automation" and "bad" are paired together. Because participants are required to provide the correct response before completing each trial, errors result in slower response times.

Scoring was completed following the *D* procedure outlined by Greenwald, Nosek, and Banaji (2003), which produces a statistic similar to Cohen's *d* and indicates the relative implicit preference for automation compared to humans. The *D* procedure was developed as a result of systematic research conducted by Greenwald and colleagues that compared several alternative scoring protocols. It was found to be the most

TABLE 2: Scale Means, Standard Deviations, and Intercorrelations

	Scale	*M*	*SD*	1	2	3	4	5
1	Propensity to trust machines	3.56	0.60	1.00				
2	Implicit attitude toward automation	−0.36	0.39	.00	1.00			
3	Trust (obviously good block)	3.55	0.70	.41**	.13	1.00		
4	Trust (ambiguous block)	3.13	0.72	.30*	.30*	.35**	1.00	
5	Trust (obviously poor block)	2.77	0.89	.04	.20	.23	.74**	1.00

Note. Pairwise *N*s = 57–59.
*$p < .05$. **$p < .01$.

appropriate option tested in regard to several factors including resistance to artifactual influences, correlations with explicit measures, and internal consistency (see Greenwald et al., 2003, for more details). This procedure now represents the standard and most widely used IAT scoring protocol.

Briefly, the *D* procedure first involves removing outlier responses that occurred in less than 300 ms or more than 10,000 ms. Then, the difference between the individual's average response time in the blocks in which "automation" was paired with "good" and the blocks in which "automation" was paired with "bad" is calculated, producing a within-person difference score. Finally, the difference score is divided by the individual's standard deviation in response times to account for individual differences in processing and response speeds. In our study, the IAT scale mean was $D = -.35$, suggesting that, on average, participants had a small to moderate implicit preference for humans over machines—in other words, the mean implicit attitude toward automation was slightly negative.

RESULTS

Scale means, standard deviations, and intercorrelations can be found in Table 2. As expected, explicit propensity to trust and implicit attitude toward automation were separate constructs—they did not correlate significantly ($r = .00$). The hypotheses and research question were tested using linear regression. Propensity to trust and implicit attitude toward automation were centered prior to entry to minimize collinearity among the predictors and their interaction term (Aiken & West, 1991). Propensity to trust and implicit attitude toward automation were entered in Block 1 of the regression, and the interaction of the two predictors was entered in Block 2. The results can be found in Table 3.

Hypothesis 1a suggested that when the automation's performance was clearly good, propensity to trust would be associated with higher actual trust. The *B* weight for propensity to trust was significant, supporting Hypothesis 1a ($B = 0.47$, $p < .01$). Neither the main effect of implicit attitude toward automation ($B = 0.22$, $p = .32$) nor the interaction ($B = 0.48$, $p = .19$) was significant. This nonsignificant result is consistent with past findings that implicit attitudes may have little effect in unambiguous situations (Dovidio & Gaertner, 2000; Fiske, 1998).

We next examined the ambiguous situation. Hypothesis 1b suggested that, in this situation, propensity to trust would again be associated with higher actual trust. Our regression indicated that this was the case ($B = 0.35$, $p < .05$), supporting Hypothesis 1b. However, unlike the previous results, participants' implicit attitudes toward automation also significantly predicted trust, such that participants with more positive implicit attitudes toward automation had higher trust ($B = 0.57$, $p < .05$). The interaction was not significant ($B = 0.37$, $p = .33$). These results imply that explicit propensity to trust and implicit attitude toward automation had additive effects such that both were associated with higher trust when automation performance was ambiguous.

TABLE 3: Linear Regressions: Hypotheses 1a–1c and Research Question 1

						95% CI		
		B	*SE*	*t*	*p*	Lower	Upper	R^2
Trust 1	Constant	2.90*	0.22	13.39	.00	2.46	3.33	
Block 1	Propensity	0.47*	0.14	3.30	.00	0.19	0.76	.15
	IAT	0.22	0.22	1.01	.32	−0.22	0.67	
Block 2	Prop × IAT	0.48	0.36	1.32	.19	−0.25	1.21	.16
Trust 2	Constant	2.63*	0.23	11.66	.00	2.18	3.08	
Block 1	Propensity	0.35*	0.15	2.38	.02	0.06	0.65	.16
	IAT	0.57*	0.23	2.49	.02	0.11	1.03	
Block 2	Prop × IAT	0.37	0.38	0.99	.33	−0.39	1.13	.16
Trust 3	Constant	2.71*	0.30	9.01	.00	2.11	3.32	
Block 1	Propensity	0.05	0.20	0.25	.81	−0.35	0.45	.00
	IAT	0.43	0.31	1.41	.16	−0.18	1.04	
Block 2	Prop × IAT	1.10*	0.49	2.27	.03	0.13	2.08	.07

Note. IAT denotes the Implicit Association Test used to assess implicit attitude toward automation. Prop × IAT denotes the multiplicative interaction of explicit propensity to trust and implicit attitude toward automation. Trust 1 is trust when the automation's performance was clearly good, Trust 2 is trust when the automation's performance was ambiguous, and Trust 3 is trust when the automation's performance was clearly poor.
*$p < .05$.

Finally, we examined the clearly poor condition. Hypothesis 1c suggested that individuals with higher propensity to trust would have lower actual trust in this situation. Contrary to Hypothesis 1c, no significant main effects were found for propensity to trust or for implicit attitude toward automation. However, a significant interaction was found ($B = 1.10$, $\Delta R^2 = .07$), as displayed in Figure 6. Those who had *both* higher explicit propensity to trust and more positive implicit attitude toward automation had higher actual trust than those who lacked one or both. Thus, the combination of high propensity to trust machines and positive implicit attitude toward automation seemed to buffer against the severe decreases in trust often seen when automation makes obvious errors.

DISCUSSION

This study examined the influence of propensity to trust and implicit attitude toward automation on trust in automation. It appears that both the explicit and implicit processes examined had significant implications for actual levels of trust. However, the relationships found differed according to the nature of the automation's performance.

The results suggest that when the system's performance was unambiguous and good, propensity to trust was associated with higher trust in the system, as found in past research. Propensity to trust was also significantly associated with trust when the automation's performance was ambiguous. This is consistent with the perfect automation schema hypothesis for the operation of propensity to trust.

When automation performance was ambiguous, users' implicit attitudes toward automation and propensity to trust combined additively to predict trust. This is the first demonstration of the potential role of implicit attitudes toward automation in influencing trust in automation. We demonstrate that an individual's implicit attitude or "gut reaction" to automation in general significantly influences trust in a specific automated system during task performance above and beyond a known explicit predictor, propensity to trust. Given that ambiguous performance is a characteristic of many applications of automation, it is important to understand

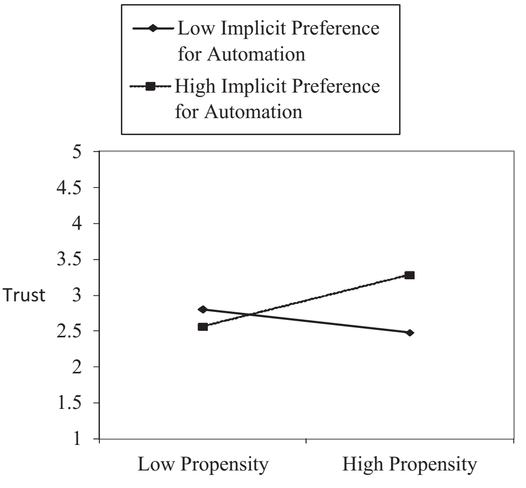

Figure 6. Interaction of propensity to trust machines and implicit preference for automation predicting trust in the clearly poor performance situation (Hypothesis 1c).

influences on trust in ambiguous situations. For example, X-ray security screeners are unlikely to learn that they missed prohibited items in luggage until minutes or hours later, if at all.

Our results suggest that implicit attitude toward automation, like explicit propensity to trust, may affect the way users perceive and interpret ambiguous performance. This suggests that implicit attitude toward automation in general could be an important lever that can be used to calibrate trust in ambiguous situations. When performance was unclear, users interpreted the information in line with their implicit attitude. Thus, interventions designed to influence implicit attitude toward automation may be effectively employed to calibrate trust levels. Such interventions are discussed further under "practical applications."

Furthermore, when obvious automation errors were encountered, an interactive effect was found such that users with both high propensity to trust and positive implicit attitudes toward automation had significantly higher actual trust than those lacking either or both. This interaction effect is particularly interesting given past findings that, compared with those with lower propensity to trust, individuals with high propensity to trust often show marked decreases in trust after an automation error (e.g., Merritt & Ilgen, 2008). Our results suggest that a positive implicit attitude toward automation combined with high propensity to trust may buffer trust against the decreases often associated with high propensity to trust alone.

This buffering may have either positive or negative implications for performance depending on the user's initial level of trust and the reliability of the specific automation in question. Nevertheless, the results suggest that individuals with high levels of propensity to trust and a positive implicit attitude toward automation may be at increased risk of complacency potential (Singh, Molloy, & Parasuraman, 1993) and increased automation misuse. Conversely, these individuals may be less likely to disuse automation. Thus, these personal characteristics might be used to select appropriate automation operators, or they might be altered via training to more effectively calibrate trust.

Our results provided support for both the additive model and the interactive model of implicit–explicit relationships. In Block 2, we found main effects for both propensity to trust and implicit preference, suggesting an additive model. However, in Block 3, no main effects were found, and instead a significant interaction was detected—suggesting an interactive model. These results are consistent with mixed results found in past work (e.g., Perugini, 2005). One possibility is that the additive model might apply in more ambiguous situations, whereas the interactive model might apply in situations where the target's behavior is clearer. However, more research is needed to test this notion.

Practical Implications

We found that user trust in automation is influenced by both implicit and explicit attitudes. Of importance, people are unaware of the effects their implicit attitudes have on their attitudes and behavior (Gawronski et al., 2006). Therefore, the results of this study imply that users do not fully understand why they experience a given level of trust in an automated system. When asked why they trust or do not trust a system, the users will be capable of describing the effects of only their explicit attitudes—not their implicit ones. Thus, interviews, surveys, and focus groups may provide an incomplete understanding of factors influencing user trust. To better understand why users trust automation, implicit attitudes must be measured using carefully crafted implicit techniques such as an IAT.

Furthermore, to correctly calibrate trust, it may be necessary to manipulate users' implicit attitudes. To do so, interventions designed to have implicit influence are necessary. Research has established that techniques useful for changing explicit attitudes, such as providing information (e.g., telling the user the automation's reliability) and the use of persuasive techniques, do not have significant effects on implicit attitudes (Gawronski & Bodenhausen, 2006; Gawronski & Strack, 2004; Rydell & McConnell, 2006). Instead, evaluative conditioning techniques may be needed. For example, Gibson (2008) found that implicit attitudes toward Coke and Pepsi could be significantly altered by repeatedly pairing one brand of soda with "good" and the other with "bad." Evaluative conditioning has also been shown to alter implicit attitudes toward the self, candy, snacks, math motivation for women, and racial prejudice (Dasgupta & Greenwald, 2001; Dijksterhuis, 2004; Ebert, Steffens, von Stülpnagel, & Jelenec, 2009; Forbes & Schmader, 2010; Hollands, Prestwich, & Marteau, 2011; Olson & Fazio, 2006).

Limitations and Suggestions for Future Research

The present study provided an initial inquiry into the role of implicit attitudes in user trust, but many questions remain. First, our results did not show a marked decrease in trust for individuals high in propensity to trust in obvious-error situations. Thus, additional research on potential moderators of that effect is needed. Second, as discussed earlier, this study measured implicit attitudes but did not attempt to alter them. Future research should examine the malleability of user implicit attitudes toward automation. Third, the present study was brief in duration, and more longitudinal research is needed on the effects of implicit attitudes over time. Fourth, ambiguity of system performance was found to be one moderator of the effects of implicit attitudes. Other moderators could be explored, including cognitive workload, motivation to control implicit attitudes, and situational constraints. In addition, all participants in our study experienced a decrease in automation reliability over the course of the task. Future research could explore the extent to which the relationships examined might generalize to other reliability scenarios.

Furthermore, context effects have been found on implicit attitudes in past work (cf. Gawronski & Bodenhausen, 2006). Although our study focused on associations with automation in general, situational moderators may exist. For example, automation in a security screening context might produce different reactions than automation in an automobile manufacturing context. Because the mental associations with automation might vary across these contexts (e.g., in the latter, automation may be associated with job insecurity), specific situational cues could prime different associations. Such context effects have been found in the domain of racial attitudes (e.g., Wittenbrink, Judd, & Park, 2001). If context effects are also relevant to automation attitudes, situational factors could be employed to influence the implicit attitudes activated and, in turn, influence explicit trust in a particular automated system. Future research could explore the effects of various contexts on implicit attitudes toward automation.

Finally, we acknowledge that our manipulation of automation performance was intertwined with task difficulty. The errors the automation made in Block 2 (ambiguous) were committed on more difficult slides than those made in Block 3. The lower difficulty of the Block 3 error slides increased participants' ability to judge the automation's performance (e.g., participants had only a 1% agreement rate on the Block 3 miss error slide). Although difficulty and observability may often be correlated in many tasks, we recommend that future research use manipulations designed to compare their unique effects.

CONCLUSION

The present study adds to a growing body of literature suggesting that many of our interactions with the world are highly driven by implicit processes (Bargh & Chartrand, 1999). We demonstrate that implicit attitudes toward automation in general can significantly affect trust in a specific automated system. Thus, implicit attitudes toward automation could have implications for reliance behavior and for safety- and performance-related outcomes. Future research should continue to examine implicit effects on interactions with automation to better understand and utilize such processes to calibrate trust and to maximize safety and performance.

APPENDIX

Propensity to Trust Scale Items

1. I usually trust machines until there is a reason not to.
2. For the most part, I distrust machines.
3. In general, I would rely on a machine to assist me.
4. My tendency to trust machines is high.
5. It is easy for me to trust machines to do their job.
6. I am likely to trust a machine even when I have little knowledge about it.

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KEY POINTS

- When automation performance was ambiguous, implicit attitude toward automation was associated with higher trust in automation.
- When automation performance was obviously good or ambiguous, propensity to trust machines was associated with higher trust in automation.
- When the automation made obvious errors, propensity to trust and implicit attitude interacted—when both were positive, users had higher trust.
- Because implicit processes may affect trust outside of awareness, users may not fully understand or be able to report why they experience a given level of trust.
- Implicit attitudes might be used as a lever for calibrating trust.

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